

DiffusionDxNet: Improving Monkeypox Classification Using Diffusion-Based Data Augmentation and DenseNet121

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Abstract—Monkeypox is an emerging infectious disease whose skin manifestations closely resemble other dermatological conditions such as chickenpox and measles, making early visual diagnosis challenging. This diagnostic similarity may delay early detection, highlighting the need for automated and reliable computer-aided diagnostic systems.

In our research, we develop a deep learning model using diffusion-based data augmentation and the DenseNet121 model to classify monkeypox into various categories. Because monkeypox image datasets are most often unbalanced, we create synthetic images of minor categories to develop a balanced dataset. This is followed by fine-tuned four-class classification tasks using a pre-trained DenseNet121 model. To make our model more robust for better predictive performance, we investigate the combination of multiple learned models through both the uniform and greedy model soup methods. The experiments were conducted using the Monkeypox Skin Images Dataset (MSID), a publicly available four-class dermatological dataset containing 770 images collected from internet-based sources and curated by the Department of Computer Science and Engineering, Islamic University, Kushtia, Bangladesh.

The proposed approach was evaluated on an independent test set using standard metrics including accuracy, precision, recall, F1-score, Matthews correlation coefficient, and ROC-AUC. The model achieved an accuracy of 97.06%, macro precision of 96.05%, macro recall of 95.35%, macro F1-score of 95.65%, and ROC-AUC of 0.9969. These results indicate consistent performance across all classes and demonstrate the effectiveness of diffusion-based augmentation combined with robust feature learning.

In conclusion, the proposed framework demonstrates strong potential as an efficient and scalable solution for automated monkeypox diagnosis and highlights the usefulness of diffusion-based augmentation in medical image analysis.

Index Terms—Monkeypox classification, Diffusion-based data augmentation, DenseNet121, Model soup, Deep learning, Medical image analysis.

I. INTRODUCTION

Monkeypox is a newly emerging viral zoonosis that has recently gained widespread attention because of the speed of its spread and the fact that its symptoms resemble existing dermatological illnesses like varicella (chickenpox) and measles. Given the fact that the symptoms of the monkeypox virus are largely visible through the lesions that it produces upon the

skin of the patient, the visual inspection of the patient is an important aspect of the diagnosis. However, diagnosis largely relies on manual clinical expertise, which may lead to delayed and inconsistent assessment [5].

There has been promising research done in recent times in the case of medical image analysis using deep learning techniques in relation to skin lesion analysis and infectious disease diagnosis. Various research works were conducted using frameworks that integrated CNN architectures in monkeypox detection in skin images [6], [9], [10]. Even though such frameworks demonstrated promising performance in monkeypox detection, their performance is normally hampered by the imbalance in monkeypox image datasets that were used in those researches.

To counteract these drawbacks, ensemble-based and feature-enhanced deep architectures have emerged in current literature. SkinMarkNet has been proposed by Akram et al. as an ensemble model encompassing data augmentation strategies to improve the accuracy of monkeypox image prediction [2]. Likewise, the need for feature extraction has been addressed by Demir et al. through their model MNPDenseNet, which employed a pretrained DenseNet model along with patch-based feature extraction for monkeypox image detection [4]. Despite these recent developments, most of the previous models are reliant on conventional data augmentation methods that generate images lacking diversity.

More recently, diffusion-based generative models have emerged as an effective data augmentation strategy by producing high-quality and semantically consistent synthetic images. These models show strong potential in mitigating class imbalance and enhancing robustness in medical image classification tasks [1].

Transfer learning using pre-trained architectures such as DenseNet has gained significant attention in medical image classification due to its efficient feature reuse and improved gradient propagation. DenseNet-based models have demonstrated strong performance in monkeypox and skin lesion classification tasks [3], [11]. Recent studies also indicate that weight-averaging strategies, commonly referred to as model

soup approaches, can enhance prediction robustness while reducing model variability without increasing computational complexity.

Motivated by these advancements, this work proposes a robust monkeypox classification framework that integrates diffusion-based data augmentation with a DenseNet121 architecture. The proposed system addresses class imbalance through diffusion-driven synthetic data generation, leverages transfer learning for effective feature extraction, and explores both uniform and greedy model soup strategies to improve predictive stability. Experiments are conducted on a publicly available monkeypox skin image dataset [12]. Unlike prior studies relying on conventional augmentation and single-model learning, the proposed DiffusionDxNet framework combines diffusion-based class balancing, DenseNet feature learning, and model soup aggregation, resulting in improved minority class representation, stable predictions, and enhanced interpretability through Grad-CAM visualization.

The key contributions from the work can be stated as follows:

- A diffusion-driven augmentation strategy is proposed to mitigate class imbalance by generating semantically consistent minority-class samples.
- A transfer learning-based DenseNet121 framework is designed for robust multi-class monkeypox skin lesion classification.
- Uniform and greedy model soup strategies are incorporated to improve prediction stability and reduce model variance.
- Comprehensive experimental evaluation and visual analysis are performed to validate robustness and generalization capability.

The remainder of this paper is organized as follows. Section II reviews related work, Section III describes the proposed methodology, Section IV presents experimental results and discussion, and Section V concludes the paper with future research directions.

II. RELATED WORK

More recent studies targeted the use of deep learning and transfer learning techniques to detect monkeypox and lesions on the skin. Most of the studies used convolutional neural networks and ensemble learning to address the issue of small data sizes or class imbalance problems. Lightweight and hybrid designs proved to be very effective with small datasets, but the performance of some custom architectures was even better than generic pre-trained deep learning models. However, the performance of transformer models proved to be suboptimal with small datasets. The most significant drawback detected in the existing literature regards the small datasets with intensive data augmentation, which are not representative or applicable to real-world scenarios to some extent. Only a few existing studies addressed the use of explainable AI. A comparative assessment of current monkeypox and skin lesion classification research, including their techniques, important contributions, and limitations, is presented in Table I.

III. METHODOLOGY

Fig. 1 illustrates the DiffusionDxNet framework, which combines a DenseNet121 classifier with diffusion-driven data augmentation.

This section describes the dataset preparation process, the diffusion-based augmentation strategy, and the DenseNet-based classification framework. The training configuration and an evaluation protocol used are also reviewed.

A. Dataset Description

Experiments were conducted using the publicly available Monkeypox Skin Image Dataset (MSID) obtained from Kaggle. The dataset contains 770 images collected from publicly accessible internet sources and comprises four classes: *Monkeypox*, *Chickenpox*, *Measles*, and *Normal* skin images. The original data distribution is highly imbalanced, with minority classes containing substantially fewer samples than majority classes, reflecting real-world medical data characteristics. Such imbalance can bias model learning toward dominant classes and negatively affect generalization performance.

B. Dataset Splitting Strategy

The dataset was split using a stratified splitting approach into training, validation, and test subsets to guarantee accurate performance evaluation. With a fixed random seed to guarantee reproducibility, the data were divided into 70% for training, 15% for validation, and 15% for testing. Crucially, in order to give an objective assessment of the model's capacity for generalization, the test set was kept entirely hidden during training and augmentation.

C. Diffusion-Based Data Augmentation

Diffusion-based data augmentation was used to rectify the substantial class imbalance in the original monkeypox skin picture dataset. A Stable Diffusion model was used to create synthetic images specifically for minority groups, guaranteeing both visual realism and semantic consistency.

Fig. 2 demonstrates that the suggested diffusion-based augmentation technique considerably improved the balancing of the class distribution, making it less biased towards the majority classes in the training stage.

To overcome class imbalance, Stable Diffusion v1.5 was employed to perform diffusion-based data augmentation. Synthetic images were created exclusively for minority classes in the training set until all classes had a nearly balanced distribution. Text-guided prompts pertaining to each skin disease category were utilized to create high-quality, realistic lesion images. This strategy increases data diversity while maintaining semantic consistency, boosting the trained model's robustness. No fake samples were included in the validation or test sets.

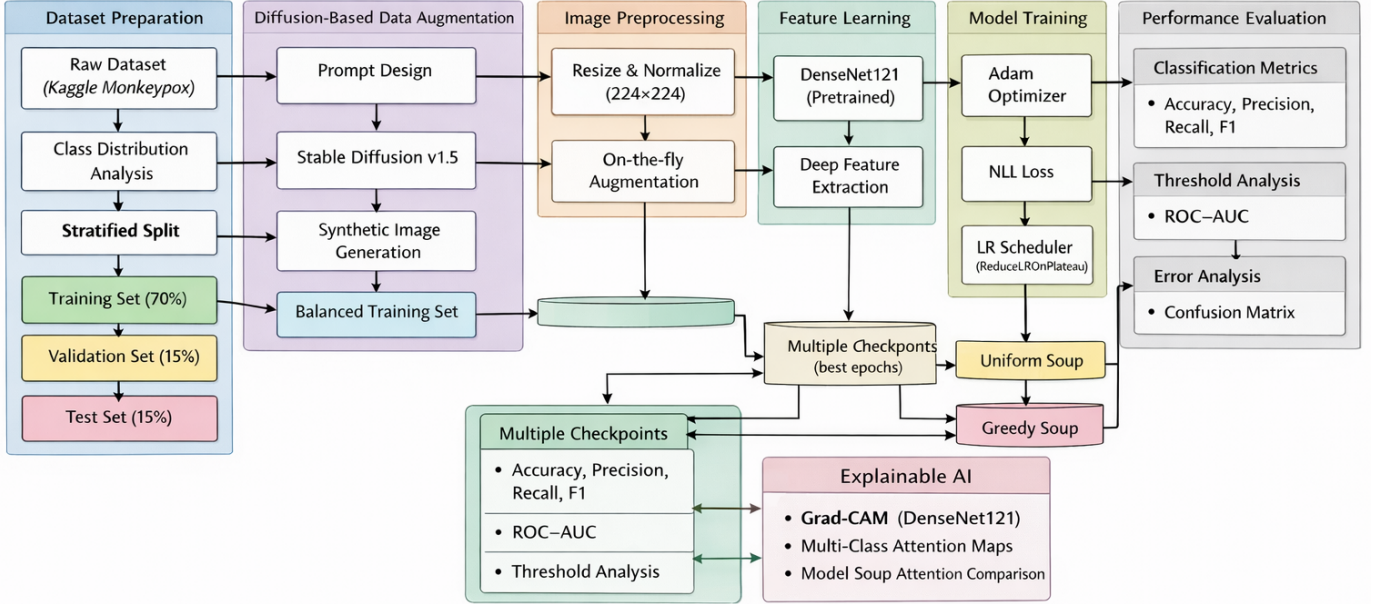
D. Image Preprocessing

The photos were reduced to 224×224 pixels to meet the deep learning model's input requirements. Pixel values were

TABLE I: Literature Review Summary

Papers	Methodology	Contribution	Limitation
Demir et al. (2024) [4]	MNPDenseNet using DenseNet201 with nested patch-based feature extraction	Achieved high accuracy in five-class skin lesion classification	Limited dataset size restricts generalization
Nayak et al. (2023) [10]	Evaluation of pretrained CNNs and Vision Transformer models	Identified ResNet-18 as the most effective architecture	Strong reliance on data augmentation
Uysal (2023) [7]	Hybrid CNN-LSTM model combining RepVGG and MnasNet	Improved generalization over individual CNN models	Limited and imbalanced dataset
Uzun Ozsahin et al. (2023) [8]	Custom 2D-CNN for binary skin lesion classification	Outperformed standard pretrained models	Extensive augmentation due to data scarcity
Yolcu Oztel (2024) [3]	Ensemble of Vision Transformer and DenseNet201	Captured both global and local lesion features	Transformer sensitivity to limited data
Altun et al. (2023) [9]	Transfer learning with optimized lightweight CNNs	Achieved high accuracy with mobile deployment feasibility	Non-standard, web-scraped dataset
Manohar and Das (2024) [2]	Ensemble learning with multiple pretrained CNNs	Improved prediction stability through ensemble strategy	Small and low-diversity dataset
Sorayaie Azar et al. (2023) [11]	DenseNet-based models with XAI integration	High accuracy with interpretability support	Very limited non-clinical dataset
Jaradat et al. (2023) [6]	Transfer learning using efficient CNN architectures	Strong performance despite minimal training data	No external clinical validation
Bathula et al. (2024) [5]	Narrative review of AI-blockchain integration	Outlined secure and explainable health-care vision	Lack of implementation-level validation
Khan et al. (2024) [1]	Bayesian inference with GLCM texture analysis	Identified discriminative feature relationships	Limited end-to-end deep learning integration

End-to-End Architecture of the Proposed DiffusionDxNet Framework



* Diffusion augmentation is performed offline prior to model training.

Fig. 1: End-to-end architecture of the proposed diffusion-augmented DenseNet121 framework for multi-class monkeypox skin lesion classification. The pipeline illustrates dataset preparation, diffusion-based data augmentation, model training, and evaluation stages.

standardized based on dataset-specific mean and standard deviation values. To boost generalization further, traditional data augmentation techniques such as random horizontal flipping and small-angle rotations were only used during training. To achieve fair evaluation, validation and test photos were processed without any augmentation.

E. DenseNet121-Based Classification Model

The backbone network was built using a DenseNet121 architecture that has been pre-trained on the ImageNet dataset. DenseNet enables effective feature reuse through dense connectivity, resulting in enhanced gradient flow and reduced overfitting. To accommodate the four target classes, the origi-

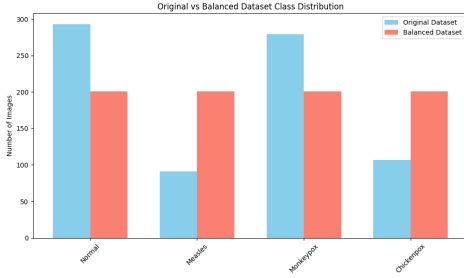


Fig. 2: A comparison of class distributions before and after diffusion-based augmentation demonstrates that class imbalances are effectively reduced.

nal classification head was replaced with a task-specific fully linked layer and a LogSoftmax activation function. Transfer learning was used to fine-tune the network parameters on the skin lesion dataset.

F. Training Configuration

The model was trained with the Adam optimizer and a learning rate of 1×10^{-4} . The negative log-likelihood loss function was used because of the LogSoftmax output layer. Training was done for 60 epochs with batch sizes of 32 and 64 in several experimental runs. A learning rate scheduler based on validation loss (ReduceLROnPlateau) was used to improve convergence stability and prevent overfitting.

G. Model Soup Strategy

To improve prediction stability and reduce model variance, a model soup technique was considered. Multiple DenseNet121 checkpoints from various training epochs and setups were combined using two methods: uniform model soup, which evenly averages model weights, and greedy model soup, which incrementally includes models based on validation performance. This method seeks to smooth the weight space while preserving the underlying architecture.

H. Performance Evaluation Metrics

Multiple metrics were used to assess model performance, including accuracy, precision, recall, macro-averaged F1-score, Matthews Correlation Coefficient (MCC), and ROC-AUC. Confusion matrices were also used to investigate class-wise prediction behavior. This thorough evaluation system enables a fair evaluation of categorization performance across all classes.

IV. RESULTS AND DISCUSSION

This section presents the experimental results of the proposed diffusion-augmented DenseNet121 framework, including quantitative evaluation and visual analysis of model performance.

A. Quantitative Performance Evaluation

The classification performance of the proposed model is evaluated on an independent test set using a variety of measures, including accuracy, precision, recall, F1-score, Matthews Correlation Coefficient (MCC), and ROC-AUC. As seen in Table II, the DenseNet121 model demonstrates strong

TABLE II: Performance comparison of DenseNet121 and model soup strategies on the test dataset

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
DenseNet121	0.9706	0.9605	0.9535	0.9565	0.9969
Uniform Soup	0.9706	0.9605	0.9535	0.9565	0.9969
Greedy Soup	0.9706	0.9605	0.9535	0.9565	0.9969

TABLE III: Comparison with Similar Monkeypox Classification Studies

Study	Model	Augmentation	Classes	Accuracy	Key Limitation
Alun et al. [9]	CNN (TL)	Conventional	4	94–96%	Web-scraped dataset
Nayak et al. [10]	ResNet-18	Conventional	4	95.0%	Small dataset
Demir et al. [4]	DenseNet201	Patch-based	5	96.8%	High model complexity
Ozcel [3]	ViT + DenseNet	Conventional	4	96.2%	Data sensitivity
Akram et al. [2]	CNN Ensemble	Conventional	4	96.5%	Limited diversity
Proposed	DenseNet121 + Soup	Diffusion-based	4	97.06%	Limited evaluation on non-clinical data

performance across all evaluation metrics. The use of uniform and greedy model soup procedures produces consistent prediction behavior and stable classification outcomes.

Notably, the equal performance scores observed between the baseline DenseNet121 model and the model soup variants show that the individual trained models converge to similar optimal solutions. This result shows that the learned feature representations are robust and well-generalized, and that weight averaging does not degrade performance.

Such stability is especially important in medical image classification applications, where consistent predictions are essential.

B. Confusion Matrix Analysis

Fig. 3 presents the confusion matrix comparison among the DenseNet121 baseline and the model soup variants. Strong diagonal dominance across all models indicates accurate class-wise prediction with minimal misclassification. The consistent confusion patterns further highlight the stability of the learned representations and confirm that diffusion-based augmentation supports balanced and reliable classification performance.

C. Training Convergence Analysis

As seen in Fig. 4, the training and validation loss values continue to decrease, confirming the model’s sustained convergence. The parallel structure of the training and validation loss curves implies that the model has a great generalization capacity and does not overfit. The above demonstrates the effectiveness of data augmentation and transfer learning via diffusion in improving training stability.

D. Learning Curve Analysis

Fig. 5 demonstrates that the proposed model achieves consistent and high validation F1-scores despite little training data. Performance variability at lower data fractions diminishes when more samples are provided, demonstrating improved learning stability.

Overall, the findings demonstrate the strong generalization capability enabled by diffusion-based augmentation, particularly in data-scarce scenarios.

E. ROC and Precision–Recall Curve Analysis

As shown in Fig. 6, the proposed model yields near-perfect ROC–AUC values for all classes, indicating significant discriminative capability between distinct skin lesion types.

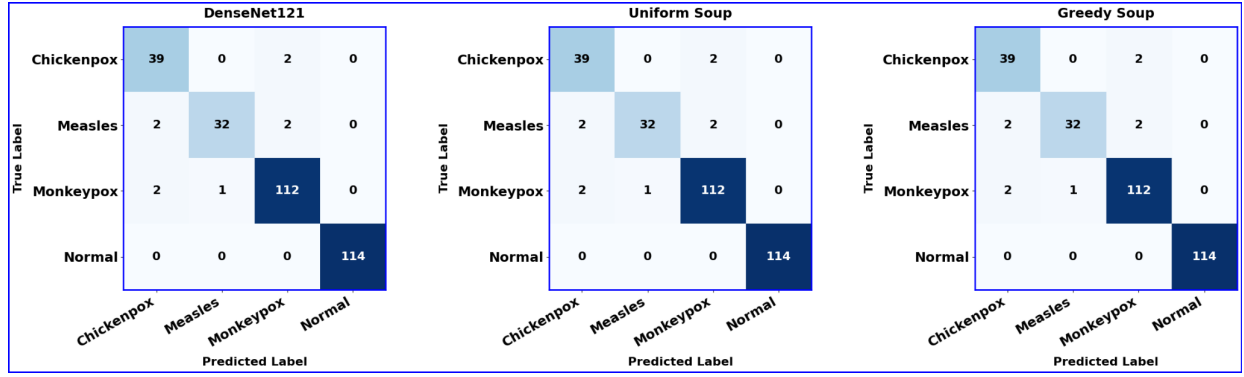


Fig. 3: Confusion matrix comparison of DenseNet121, uniform model soup, and greedy model soup approaches on the test dataset. Strong diagonal dominance across all models indicates accurate and consistent class-wise prediction performance.



Fig. 4: The training and validation loss curves exhibit a smooth and steady fall, demonstrating stable convergence without overfitting.

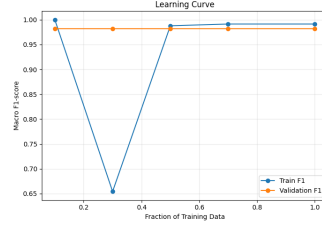


Fig. 5: The learning curve depicts the macro F1-score trends of the proposed diffusion-augmented DenseNet121 model as training data percentages increase.

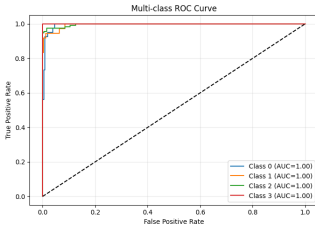


Fig. 6: Multi-class ROC curves of the proposed model.

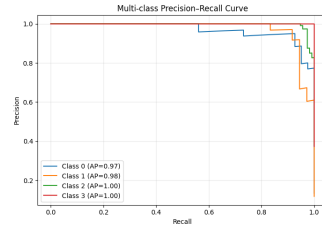


Fig. 7: Multi-class Precision-Recall curves of the proposed model.

The ROC curves are continuously near to the top left corner, indicating high true positive rates and low false positives.

Fig. 7 demonstrates the usefulness of the suggested framework through precision-recall analysis. The high average precision values across all classes demonstrate that the model can retain reliable precision even at high recall levels. This tendency is especially relevant in medical diagnostics, where correct identification of disease-positive instances is essential.

F. Impact of Diffusion-Based Data Augmentation

The use of diffusion-based synthetic image synthesis is critical in addressing class imbalance in the training dataset. By supplementing minority classes with high-quality and semantically consistent samples, the model gains feature diversity and reduces bias toward majority classes. This improvement

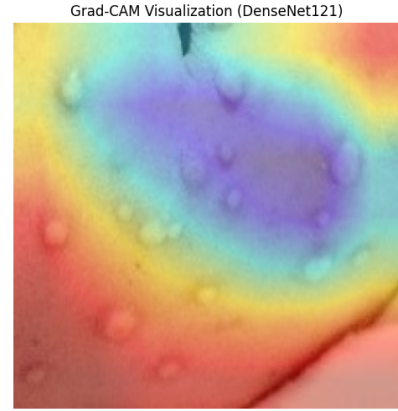


Fig. 8: The Grad-CAM results indicate that DenseNet121 concentrates on clinically relevant lesion regions associated with monkeypox, demonstrating meaningful attention to discriminative skin patterns and supporting the interpretability of the model's predictions.

is seen in the balanced performance indicators and consistent confusion matrix results across all categories.

Overall, the experimental results show that combining diffusion-based augmentation with DenseNet121 and model soup techniques creates a strong and effective framework for automated monkeypox classification.

G. Grad-CAM Based Model Interpretability

To increase the transparency of the proposed DiffusionDxNet framework, Gradient-weighted Class Activation Mapping (Grad-CAM) is used to depict class-discriminative regions learned by the DenseNet121 backbone. Grad-CAM identifies spatial regions in the input image that provide the greatest contribution to the model's prediction by using gradient information from the final convolutional layers.

Fig. 8 shows the Grad-CAM activation map emphasizing clinically relevant lesion regions guiding the model prediction. The highlighted areas indicate that the network prioritizes disease-specific patterns rather than background artifacts, supporting the reliability of the learned representations, suggesting that the model prioritizes genuine diseased features over background artifacts. This behavior demonstrates the

learned feature representations' dependability and therapeutic relevance.

Overall, the Grad-CAM analysis reveals that the proposed model bases judgments on localized lesion-specific cues, which improves the diagnostic framework's interpretability and trustworthiness without changing its main learning pipeline.

These findings highlight that diffusion-based augmentation contributes not only to class balance but also to improved feature diversity, which enhances generalization capability under limited data conditions. The consistent performance across baseline and model soup variants suggests that the learned representations are stable and less sensitive to weight perturbations. Furthermore, Grad-CAM analysis confirms that the model focuses on clinically meaningful lesion regions, strengthening the reliability of the proposed diagnostic framework.

V. FUTURE WORK

Although the proposed diffusion-augmented DenseNet architecture works well and robustly, many areas remain to be investigated. The primary extension would be to carry out extensive clinical validation with heterogeneous datasets collected from many institutions and imaging scenarios to assess robustness against variability in the real world.

Even though explanations based on Grad-CAM have been employed to offer a visual understanding of class discriminatory regions, in future studies, it may be worthwhile to investigate the application of more sophisticated methods of explainable AI, such as counterfactual explanations and concept-based explanations.

On the other hand, in terms of research methodology, diffusion-based data augmentation techniques can be employed in class-aware sampling/reweighting approaches to enhance the minority class. Additionally, generalizing the proposed framework to encompass other forms of dermatological/infectious skin disorders can further cement it as a generic diagnostic tool.

VI. CONCLUSION

This study presents a computer-aided framework for monkeypox skin lesion classification by integrating diffusion-based data augmentation with DenseNet121 feature learning. The proposed approach effectively addresses the challenges of limited and imbalanced medical image datasets, enabling robust and reliable multi-class prediction.

The experiments showed that the diffusion-based augmentation balanced class representation and improved generalization while DenseNet121 turned out to be effective in extracting features relevant for the discrimination of multiple classes of skin lesions. Moreover, the alignment between the baseline model and model soup approaches evidenced that the learned representations are stable, a property much welcomed in the construction of reliable medical decision-support systems. However, the study is limited by the use of non-clinical web-sourced images, motivating future validation on real-world clinical datasets.

In general, the results show that the diffusion-augmented deep learning framework is promising in automated monkeypox diagnosis and has strong potential to support early detection in the clinical setting.

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